

Developer Guide — Decoder

From first project to a fully featured DCC decoder

Covers the library on any platform; the TI MSPM0G3507 LaunchPad is the worked example

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1. Introduction

OpenDccCLib is a C library that implements the NMRA DCC protocol for microcontrollers. This guide covers the decoder role: receiving DCC from the track, recognizing the commands addressed to this decoder, and acting on speed, function, accessory, and CV programming commands. It allocates no dynamic memory and needs no operating system; a main loop and one edge interrupt are enough.

1.1 What the Library Does (and Does Not Do)

The library classifies one and zero bits from edge timing, assembles packets and validates the XOR byte, matches locomotive and accessory addresses against the CVs, dispatches commands to your callbacks from the main loop, wraps CV storage with the decoder lock and factory reset, answers service-mode programming with the 6 ms ACK pulse, runs the S-9.2.4 packet time-out, and, when wired, transmits RailCom replies during the cutout.

It does not include motor control, lighting, servos, sound, or persistent storage. You supply those through function pointers in one `dcc_config_t` struct, and the library calls them.

1.2 Platform Support

Processor / Board	Example IDE / Toolchain
TI MSPM0G3507 LaunchPad	Code Composer Studio / Theia (the shipped example)
RP2040 / RP2350	Arduino IDE, PlatformIO, Pico SDK
STM32 (F4 and others)	STM32CubeIDE
ESP32	Arduino IDE, PlatformIO
Any ARM Cortex-M	GCC + Makefile

The only hardware a decoder needs is a GPIO edge interrupt and a microsecond timestamp. RailCom transmit additionally needs a delay accurate to 1 μ s, and the edge interrupt must then call the library directly.

2. DCC Decoder Concepts

2.1 Signal Reception

The track carries a bipolar square wave. The decoder rectifies it for power and converts it to a logic-level signal through an optocoupler or divider. Every transition is a half-bit boundary; the decoder measures the time between transitions.

2.2 Bit Classification

Interval between edges	Classification	Define (dcc_defines.h)
< 80 µs	One-bit half (58 µs nominal)	DCC_DECODER_HALF_BIT_THRESHOLD_US = 80
80 µs ... 10 000 µs	Zero-bit half (≥ 95 µs; NMRA allows stretching)	
$\geq 10\,000$ µs	No signal; the decoder resets to preamble search	DCC_DECODER_HALF_BIT_MAX_US = 10000

Two consecutive halves of the same class make one bit. A short half followed by a long one does not pair; the long half starts a new pair.

2.3 Packet Structure

A valid preamble is at least `DCC_PREAMBLE_BITS_DECODER_MIN = 10` one-bits (command stations send 14 or more). A zero start bit begins each byte; a one end bit closes the packet. The last byte is the XOR of all preceding bytes; a mismatch discards the packet. Packets shorter than 2 bytes or longer than `DCC_PACKET_MAX_BYTES = 6` are discarded, and with RailCom compiled in the end bit is not counted toward the next preamble.

2.4 Address Matching

The library reads its addressing CVs at `DccConfig_initialize()` and again whenever a DCC packet writes one of them (CV 1, 8, 17, 18, 29, 513, 521, 541). Writes made through `DccApplicationDecoderCv_write` refresh it as well. If the application writes its storage directly, call `DccConfig_reload_address_cvs()` afterwards, as the demo's ADDR command does.

CV	Purpose
CV 1	Primary (short) address, 1–127
CV 17–18	Extended (long) address, 128–10239
CV 29 bit 5	0 = use CV 1, 1 = use CV 17–18
CV 19	Advanced consist address (bits 0–6, 0 = none) and direction bit 7; speed, direction and emergency-stop packets to this address are accepted, and function packets are, subject to CV 21/22
CV 21, 22	Which functions answer the consist address: CV 21 bits 0–7 for F1–F8; CV 22 bit 0 for FL travelling forward, bit 1 for FL in reverse, bits 2–5 for F9–F12. F13 and above never answer the consist address
CV 513, 521	Accessory decoder address: low 6 bits and high 3 bits in decoder-address mode; in output-address mode (CV 541 bit 6) the flat address is $\text{CV } 513 + 256 \times \text{CV } 521 - 1$

CV	Purpose
CV 541	Accessory configuration: bit 7 accessory decoder, bit 6 output-address mode, bit 5 extended

Broadcast (address 0) is always accepted for multifunction packets. A decoder configured as an accessory decoder (CV 541 bit 7) ignores multifunction packets altogether, broadcast included, so a locomotive packet whose short address equals the board address is never mistaken for a command. Accessory packets are matched against the address from CV 513/521, as a board address or, with CV 541 bit 6, as an output address; S-9.2.1 defines no accessory broadcast, so none is accepted.

2.5 Configuration Variables

CV	Name	Note
1	Primary address	Demo default 3
2–6	Vstart, acceleration, deceleration, Vhigh, Vmid	Application use
7	Manufacturer version	Read-only per S-9.2.2; the library does not enforce it, so refuse the write in <code>cv_write</code>
8	Manufacturer ID	Read-only per S-9.2.2; writing the value 8 triggers <code>factory_reset</code> (bypassing the decoder lock); any other value is forwarded to <code>cv_write</code> , so refuse it there
11	Packet time-out	Fail-safe, in 100 ms units; 0 disables
15 / 16	Decoder lock	Writes are refused unless CV 15 equals CV 16
17–18	Extended address	
19	Consist address	Written by the consist set/clear instruction; bit 7 reverses direction within the consist
21, 22	Consist function enables	Per-function: 1 = also answers the consist address, 0 = locomotive address only. CV 22 bits 0/1 are FL forward/reverse, bits 2–5 are F9–F12
28	RailCom configuration	bit 0 Channel 1, bit 1 Channel 2; not read by the library in this release, the transmit engine always sends Channel 1
29	Configuration	See <code>DCC_CV29_*_BIT</code> ; bit 6 is reserved and forced to 0
31 / 32	Indexed CV page	Selects the page behind CVs 257–512
257–512	Indexed window	Served by <code>cv_read_indexed</code> / <code>cv_write_indexed</code> when wired

3. Project File Structure

decoder/	<- your project folder
decoder.c	<- main entry: config struct, edge ISR, ring buffer, main loop
dcc_user_config.h	<- REQUIRED: role flags and limits
decoder_command_parser.c/h	<- demo UART command-line interface
application_callbacks/	
callbacks_dcc.c/h	<- CV storage stub, CV29 feature mask, command callbacks
application_drivers/	
ti_driverlib_dcc_driver.c/h	<- timestamp, lock, RailCom delay
ti_driverlib_uart_driver.c/h	<- UART I/O
ack_pulse_driver.c/h	<- ACK current load
dcc_lib/ -> ../../../../src/dcc	<- library core (symlink; do not edit)
dcc_config.h/c	- config struct, lifecycle, wiring of every module
dcc_bit_decoder.h/c	- edge timing -> bits -> bytes
dcc_packet_decoder.h/c	- XOR, address match, dispatch queue
dcc_cv_storage.h/c	- CV access, decoder lock, factory reset, indexed CVs
dcc_failsafe.h/c	- CV 11 packet time-out
dcc_railcom_decoder.h/c	- RailCom transmit engine (bit-bang)
dcc_railcom_utilities.h/c	- 4/8 code words
dcc_application_decoder_cv.h/c	- CV read/write API
dcc_application_decoder_railcom.h/c	- Channel 2 reply builders for on_railcom_request

4. dcc_user_config.h in Depth

```
#define DCC_COMPILE_DECODER
#define DCC_COMPILE_RAILCOM // RailCom transmit; comment out to strip
// #define DCC_COMPILE_ACCESSORY_DECODER // accessory-decoder RailCom helpers; needs DCC_COMPILE_RA

#define USER_DEFINED_DCC_DECODER_PACKET_QUEUE_DEPTH 8 // deferred dispatch queue, >= 2
```

Define	Meaning
DCC_COMPILE_DECODER	Bit decoder, packet decoder, CV storage, fail-safe
DCC_COMPILE_RAILCOM	Compiles the RailCom transmit engine and its config fields (railcom_tx_pin_set, railcom_delay_us, on_railcom_request); active only when railcom_tx_pin_set is wired
DCC_COMPILE_ACCESSORY_DECODER	Reserved for the accessory-decoder RailCom feature (SRQ, status, time, error), which is not implemented yet. Today it only enables the RailCom 4/8 encoders and satisfies the role check; packet reception lives under DCC_COMPILE_DECODER
USER_DEFINED_DCC_DECODER_PACKET_QUEUE_DEPTH	Packets that can wait for main-loop dispatch; one slot is reserved, so ≥ 2 and the queue holds depth – 1. A full queue drops the newest packet

A decoder project must not define DCC_COMPILE_COMMAND_STATION unless it really is both, as a booster or repeater would be.

5. Initialization — decoder.c

5.1 The dcc_config_t Struct (Decoder Fields)

Field	Required?	Purpose
lock_shared_resources, unlock_shared_resources, get_timestamp_usec	REQUIRED	Platform: interrupt mask and microsecond clock
cv_read(cv, *value), cv_write(cv, value)	REQUIRED	Persistent CV storage
cv29_apply_supported_features(flags)	REQUIRED by contract (NULL is tolerated)	On every CV 29 write, clear the feature bits this product does not implement; the library stores what you leave set
factory_reset()	optional	Called when the value 8 is written to CV 8; bypasses the decoder lock
cv_read_indexed, cv_write_indexed	optional	Serve CVs 257–512 for the page in CV 31:32
railcom_tx_pin_set(high)	optional (DCC_COMPILE_RAILCOM only)	Drive the RailCom current source. NULL = no RailCom transmit
railcom_delay_us(us)	required with the above	Busy-wait accurate to 1 µs; the library calls it with 4 (bit), 33 (gap) and 80 (blanking) µs
on_railcom_request(instruction, count, *out)	optional	Fill a Channel 2 reply for a command addressed to this decoder; return DATA, ACK, BUSY, NACK or NONE. Called on the edge path, not from the main loop; keep it fast
start_ack_pulse(), stop_ack_pulse()	optional	Service-mode ACK current load; the library times the 6 ms from DccConfig_run(). A second ACK while the pulse is active is ignored, not restarted; with start_ack_pulse NULL no pulse is ever armed
on_*_command(...), on_nfailsafe_entered/exited	optional	Application notifications, main loop

```
const dcc_config_t dcc_config = {
    .lock_shared_resources = &TI_DccDriver_lock_shared_resources,
    .unlock_shared_resources = &TI_DccDriver_unlock_shared_resources,
    .get_timestamp_usec = &TI_DccDriver_get_timestamp_usec,

    .cv_read = &CallbacksDcc_cv_read, /* RAM stub in the demo */
    .cv_write = &CallbacksDcc_cv_write,
    .factory_reset = &CallbacksDcc_factory_reset,
    .cv_read_indexed = &CallbacksDcc_cv_read_indexed,
    .cv_write_indexed = &CallbacksDcc_cv_write_indexed,
    .cv29_apply_supported_features = &CallbacksDcc_cv29_apply_supported_features,

    .railcom_tx_pin_set = NULL, /* no current source on the demo board */
    .railcom_delay_us = &TI_DccDriver_railcom_delay_us, /* uses DELAY_TIMER_INST, which must be added in Sys

    .start_ack_pulse = &AckPulseDriver_start,
    .stop_ack_pulse = &AckPulseDriver_stop,

    .on_speed_command = &CallbacksDcc_on_speed_command,
```

```

/* ... the other command callbacks ... */
.on_failsafe_entered = &CallbacksDcc_on_failsafe_entered,
.on_failsafe_exited  = &CallbacksDcc_on_failsafe_exited,
};

```

5.2 Edge Timestamp Ring Buffer

The GPIO interrupt only captures a timestamp into a power-of-two ring; the main loop drains it into `DccConfig_decoder_edge_isr()`. That keeps the interrupt under a microsecond so it never overruns the 58 μ s half-bit, and it means bit decoding runs in the main loop on the demo. Calling `DccConfig_decoder_edge_isr()` directly from the interrupt is equally valid on a faster part, and it is required once RailCom transmit is wired: the reply is timed from the end-bit edge, and a main-loop drain delays it past the cutout.

```

#define EDGE_BUF_SIZE 256 /* power of two */
static volatile uint32_t _edge_buf[EDGE_BUF_SIZE];
static volatile uint16_t _edge_head, _edge_tail;

void GROUP1_IRQHandler(void) { /* PB1 main, PB4 service */
    bool track_sel = /* PB17 reads high */;
    if ((!track_sel && /* edge on PB1 */) || (track_sel && /* edge on PB4 */)) {
        uint16_t next = (_edge_head + 1) & (EDGE_BUF_SIZE - 1);
        if (next != _edge_tail) { /* ring full: the edge is dropped */
            _edge_buf[_edge_head] = TI_DccDriver_get_timestamp_usec();
            _edge_head = next;
        }
        /* toggle PB3 (logic-analyzer aid) */
    }
    /* clear the interrupt flags */
}

static void _drain_edge_buffer(void) {
    while (_edge_tail != _edge_head) {
        DccConfig_decoder_edge_isr(_edge_buf[_edge_tail]);
        _edge_tail = (_edge_tail + 1) & (EDGE_BUF_SIZE - 1);
    }
}

```

5.3 Setup and Main Loop

```

int main(void) {
    SYSCFG_DL_init();
    NVIC_EnableIRQ(GPIOB_INT_IRQn);
    NVIC_EnableIRQ(TIMESTAMP_TIMER_INST_INT_IRQn); /* timestamp overflow counter */
    NVIC_EnableIRQ(ACK_PULSE_TIMER_INST_INT_IRQn);
    TI_DccDriver_initialize();
    TI_UartDriver_initialize();
    AckPulseDriver_initialize();
    CallbacksDcc_initialize(); /* CV defaults: CV1 = 3, CV29 = 0x06 */
    DccConfig_initialize(&dcc_config); /* reads the addressing CVs */
    DecoderCommandParser_initialize();

    while (1) {
        _drain_edge_buffer(); /* edges -> bit decoder */
        DccConfig_run(); /* packet dispatch, callbacks, ACK timing, fail-safe */
        CallbacksDcc_drain(); /* RECV lines -> UART */
        TI_UartDriver_echo_process();
        DecoderCommandParser_process();
    }
}

```

If the ring buffer overflows, edges are dropped silently and packets are lost. Keep every call in the loop non-blocking.

6. ISR Architecture

6.1 GPIO Edge Interrupt

Fires on every rising and falling edge of the DCC input. The demo has two inputs on one port, PB1 for the main track and PB4 for the service track, and a track-select input on PB17 chooses which one is accepted. During a RailCom transmit the library masks this interrupt through `lock_shared_resources`, so the decoder's own current pulses cannot trigger it.

6.2 100 ms Tick

The library's `DccConfig_100ms_timer_tick()` belongs to the command-station role; the decoder needs no periodic tick. The demo's SysTick only blinks the heartbeat LED. Fail-safe timing uses `get_timestamp_usec` from `DccConfig_run()`.

7. Implementing the Drivers

Function	Context	Contract
lock_shared_resources / unlock_shared_resources	any	Mask and restore interrupts; must cover the DCC edge interrupt
get_timestamp_usec	any	Free-running microsecond counter; monotonic, wrap is fine. Called on the edge path and from <code>DccConfig_run()</code> (fail-safe and ACK timing)
cv_read / cv_write	main loop	Persistent storage. The demo uses a RAM array that resets at power-up; use Flash or EEPROM in a product
start_ack_pulse / stop_ack_pulse	main loop	Switch the ACK current load; the library calls stop after <code>DCC_ACK_PULSE_DURATION_US = 6000 µs</code> , and ignores a start while the pulse is active
railcom_tx_pin_set	edge path (ISR when the interrupt calls the library directly)	Set the RailCom line to a UART level, true = mark
railcom_delay_us	edge path	Spin that many microseconds, accurate to 1 µs; called with 4, 33 and 80 (a hardware timer or DWT cycle counter, for example)

8. The Bit Decoder

`DccConfig_decoder_edge_isr(timestamp)` computes the interval since the previous edge and classifies it as in section 2.2. Halves pair into bits; ten or more consecutive ones arm the preamble; a zero then starts byte assembly. The finished packet is handed to the packet decoder's queue through an interface pointer, never by direct call; with RailCom compiled in it first passes through the RailCom transmit hook, and each completed byte also goes to the RailCom recognizer so the reply can be prepared before the packet ends.

9. The Packet Decoder

Completed packets are queued (`USER_DEFINED_DCC_DECODER_PACKET_QUEUE_DEPTH`) and dispatched from `DccConfig_run()`. This deferral keeps long handlers such as CV writes out of the edge path, and it is why every command callback runs in main-loop context (`on_railcom_request` is the exception; it runs on the edge path). For each packet the decoder validates the XOR, matches the address (its own, broadcast, or the consist address in CV 19 for speed, direction, emergency stop and the CV 21/22-enabled functions), then parses the instruction and calls the matching callback. A multifunction packet addressed to this decoder, or broadcast, also re-arms the fail-safe timer; accessory, service-mode, idle and reset packets do not. The reply engine's address is updated whenever the address-CV cache is refreshed, not per packet. A full queue drops the newest packet.

Instruction	Callback
Speed 14/28/128	<code>on_speed_command(address, speed, direction, mode)</code> ; mode is a <code>dcc_speed_mode_enum</code> . direction already has CV 29 bit 0 applied, and CV 19 bit 7 for a packet that arrived on the consist address; speed is 0 (stop) or 2 and up (e-stop goes to the next row), and 14 versus 28 steps follows CV 29 bit 1
Emergency stop	<code>on_emergency_stop_command(address)</code>
Functions F0–F68	<code>on_function_command(address, function_number, state)</code> ; at the consist address only the functions enabled in CV 21/22 are delivered (FL by the direction last reported to <code>on_speed_command</code>), F13 and above never
Basic / extended accessory	<code>on_accessory_basic_command(board, pair, activate)</code> , <code>on_accessory_extended_command(address, aspect)</code> ; delivered only for packets that match this decoder's accessory address. In output-address mode (CV 541 bit 6) board is the 11-bit output address and pair is the R bit. The extended address is always the 11-bit packet address A10..A0 (S-9.2.1 2.4.2); in decoder-address mode all four A1A0 values of the board are delivered
CV write / verify / bit, main track or service track	<code>on_cv_write_command(cv, value, service_mode)</code> , <code>on_cv_verify_command(...)</code> , <code>on_cv_bit_command(cv, bit, value, service_mode)</code> . An operations-mode verify only notifies (nothing is read or compared); a service-mode bit write fires both the write and the bit callback; accessory operations-mode CV access is not delivered in this release
Consist	<code>on_consist_command(address, consist_address, direction_normal)</code> ; fired after CV 19 has been written through <code>cv_write</code> (0 = cleared), and not fired when the decoder lock refuses the write
Binary state short / long	<code>on_binary_state_short_command(address, state, active)</code> , <code>on_binary_state_long_command(...)</code>
Analog function	<code>on_analog_function_command(address, output, value)</code>

10. CV Storage, Lock, and CV 29

All CV traffic goes through `dcc_cv_storage`, which wraps your `cv_read / cv_write`:

- **Decoder lock.** A write is refused unless CV 15 equals CV 16; CV 15 and CV 16 themselves are always writable. If CV 15 or CV 16 cannot be read the decoder is treated as unlocked.
- **Factory reset.** A write of the value 8 to CV 8 calls `factory_reset()` instead of storing, and bypasses the lock; any other value is forwarded to `cv_write`.
- **Indexed CVs.** CVs 257–512 are routed to `cv_read_indexed / cv_write_indexed` with the page from CV 31:32, when those hooks are wired.
- **CV 29 feature mask.** On every CV 29 write the library forces the reserved bit 6 clear, decodes the byte into a `dcc_cv29_flags_t` (direction reversed, 28/128 steps, analog conversion, RailCom, speed table, extended address, accessory), and calls `cv29_apply_supported_features()`. Clear the flags your product does not implement; per S-9.2.2 an unsupported feature bit must never be settable, and only the application knows what it supports.

The application API is `DccApplicationDecoderCv_read, _write` and `_is_locked`, wired by `DccConfig_initialize()` onto the storage module, so the decoder lock, the CV 29 filter and the CV 8 reset apply to application writes too. It behaves exactly like a write arriving by DCC packet: the decoder lock applies with its CV 15/16 and CV 8 exceptions, so a locked decoder can be unlocked or reset through the API, and both paths refresh the address cache. For storage written outside the library, call `DccConfig_reload_address_cvs()`.

11. Service Mode (Decoder Side)

On the programming track the library recognizes the service-mode sequences and handles them: a write stores through `cv_write` and answers with the ACK pulse; a verify compares through `cv_read` and answers only on a match; bit operations do the same at bit level. Your CV callbacks fire afterwards with `service_mode` true, so you can tell programming-track writes from operations-mode ones.

Service mode is entered after 3 consecutive reset packets and left on the next packet that is neither idle nor a service-mode packet. Direct-mode packets reach CVs 1–1024. The 3-byte register and paged forms map registers 1–8 straight onto CVs 1–8: the page register (6) is stored as CV 6 and the page is not applied, and register 5 lands on CV 5 rather than CV 29, so only CVs 1–4 are reached correctly through those modes.

12. RailCom Replies

With `railcom_tx_pin_set` and `railcom_delay_us` wired, the library transmits during the cutout that follows a packet addressed to this decoder, bit-banging 4/8-encoded bytes at 250 kbaud with `DCC_RAILCOM_TX_BIT_US = 4 µs` bits, after a `DCC_RAILCOM_TX_BLANK_US = 80 µs` blanking delay. Interrupts are masked for the transmit window through `lock_shared_resources`. The reply is timed from the end-bit edge, so the edge interrupt must call `DccConfig_decoder_edge_isr()` directly rather than through a main-loop drain.

- **Channel 1** carries the address broadcast, alternating ADR1 and ADR2, sent automatically after a multifunction command addressed to this decoder whose length the recognizer can size. Broadcast and accessory packets, decoder control, XPOM, time and date, and system time get no reply. CV 28 and CV 29 bit 3 are not consulted.
- **Channel 2** is yours: `on_railcom_request()` is called as soon as a command addressed to this decoder is recognized, before its XOR byte, and returns a reply status. Return `DCC_RAILCOM_REPLY_DATA` with a `dcc_railcom_response_t` (the id and the first byte share the first two code words, each further byte carries 6 bits, and the encoded reply is limited to six code words, so at most five data bytes), or ACK, BUSY, NACK, or NONE.
- The builders in `dcc_application_decoder_railcom.h` (`DccApplicationDecoderRailcom_pom_response`, `_dynamic_data`, `_cv_auto_transfer`, `_raw`) fill the out response inside `on_railcom_request`, so a callback body is one builder call and a `DCC_RAILCOM_REPLY_DATA` return. They are stateless: nothing to initialize or wire. ACK, NACK and BUSY are reply statuses, not builders.
- Accessory-decoder RailCom (SRQ, status, time and error reports) is not implemented: the transmit engine does not answer accessory packets. It is recorded as a future feature to be built together with the command-station polling side.

The shipped decoder board has no current-source circuit, so the demo leaves `railcom_tx_pin_set` NULL and no reply is transmitted. The engine and encoders are unit-tested; the on-track side is a known open item, and the compliance overview's RailCom entries predate the transmit engine. Known limitation: a reply carrying only Channel 2 data is read as Channel 1 by this library's own command station.

13. Fail-Safe

S-9.2.4 requires a decoder to stop everything when no packet addressed to it arrives within its time-out. The library implements this in `dcc_failsafe`: CV 11 times `DCC_FAILSAFE_CV11_UNIT_US = 100 000 µs`, polled from `DccConfig_run()`, stamped at initialization and re-armed by every multifunction packet for this decoder or broadcast (accessory and service-mode packets do not count). Once tripped it stays tripped until such a packet arrives. CV 11 = 0 disables it. `on_failsafe_entered()` is where you stop the motor and drop the outputs; `on_failsafe_exited()` fires when an addressed packet returns. A command station built on this library paces its refresh so an idle locomotive may go up to 120 packet cycles without an addressed packet (about 0.8 s, 1.8 s with worst-case packets), so on such a layout CV 11 should be 0 or at least the station's `DCC_REFRESH_CV11_FLOOR`, 20 (2.0 s) by default.

14. Porting to a New MCU

1. Copy the decoder example project.
2. Rewrite `ti_driverlib_dcc_driver.c/h`: interrupt mask, microsecond timestamp, and a 1 μ s-accurate `railcom_delay_us` if you transmit RailCom.
3. Replace `ack_pulse_driver.c/h` with your current-load control, or leave both ACK hooks NULL.
4. Replace the CV stub with Flash or EEPROM access.
5. Wire the edge interrupt to capture timestamps, and the ring-buffer drain (or, with RailCom, the interrupt itself) to `DccConfig_decoder_edge_isr()`.
6. Build and test against a known-good command station; the RECV lines over UART show every decoded command.

15. Unit Testing

Decoder-role tests, GoogleTest with mocked drivers, run with the rest of the suite: `cd test && make`. At generation time the whole suite is 28 binaries, 1249 tests, 0 failures, with 99.5 % line coverage.

Test file	What it tests
<code>dcc_bit_decoder_Test</code>	Edge classification, thresholds, preamble, noise and time-out recovery
<code>dcc_packet_decoder_Test</code>	XOR, address matching for every address type including the consist address, CV 19 set/clear, CV 21/22 consist function gating, instruction dispatch, address-cache reload, queue
<code>dcc_cv_storage_Test</code> , <code>dcc_application_decoder_cv_Test</code>	Lock, factory reset, indexed CVs, CV 29 feature mask
<code>dcc_failsafe_Test</code>	CV 11 time-out, enter and exit
<code>dcc_config_Test</code>	Wiring and lifecycle, edge dispatch on run, the 6 ms ACK pulse (auto-stop, restart ignored, NULL hooks), the CV application API routed through storage and the lock
<code>dcc_railcom_decoder_Test</code> , <code>dcc_railcom_utilities_Test</code> , <code>dcc_application_decoder_railcom_Test</code>	Reply engine, 4/8 encoding against the spec table, reply builders

16. Troubleshooting

Symptom	Likely cause and fix
No packets decoded	Input wiring, edge interrupt not enabled, or the ring buffer is not drained
Packets decode but callbacks stay silent	Address mismatch; check CV 1 and CV 29 bit 5, or try the broadcast address
Frequent XOR failures	Noisy input; clean the signal to logic levels and check the threshold circuit
Command station reports NO ACK	ACK current load not wired or too weak; verify a 6 ms pulse on the ACK pin
Fail-safe trips at once	No DCC reaching the decoder, or CV 11 set very small (units of 100 ms; keep it 0 or at least 20 with a refresh-pacing command station)
Functions work but speed does not	CV 29 bit 1 speed-step setting does not match the command station
Address lost after power cycle	CV storage is RAM in the demo; use non-volatile storage
No RailCom reply seen	<code>railcom_tx_pin_set</code> is NULL, <code>railcom_delay_us</code> is inaccurate, edges are drained from the main loop instead of the interrupt, the packet was not addressed to this decoder, or the instruction is one the recognizer cannot size

Library source: github.com/JimKueneman/OpenDccCLib. NMRA DCC standards: www.nmra.org/dcc.